

EchoVision: A Handheld Mixed Reality Mask for Experiencing Bat Echolocation

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Figure 1: EchoVision is a handheld mixed reality mask to simulate bat echolocation.

ABSTRACT

What is it like to be a bat? "EchoVision" is a mixed reality interactive experience that immerses participants in the world of bats through sound visualization and mixed reality technology. With a custom-designed, handheld, bat-shaped, optical see-through mixed reality mask based on the open-source HoloKit project, participants can simulate echolocation, the natural navigation method bats use in the dark. Participants use their voices and interpret the returned echoes. The visual feedback is dynamically adjusted based on the user's voice pitch and tone, as well as the 3D shape of the surrounding environment detected by LiDAR on the mask, offering an interactive and dynamic depiction of how bats perceive their surroundings.

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Compared to other mixed reality headsets, the pop-up exhibition in the bat habitat demonstrates that this unique mask design can effectively accommodate a large volume of audiences in-the-wild, offering them scientific education with empathetic engagement. This work promotes an ecocentric design perspective and fosters understanding between species, educating and cultivating a deeper appreciation for the unique ways non-human creatures engage with their ecosystems.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing devices**; **Ubiquitous and mobile devices**; **Mixed / augmented reality**.

KEYWORDS

Handheld Mixed Reality Mask, Ecocentric design, Non-Anthropocentric design, Augmented Reality, Echolocation Visualization

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1 INTRODUCTION

People often harbor misconceptions about bats, viewing them as dangerous, dirty, or associated with negative superstitions. Common myths include the belief that bats are blind, that they commonly get tangled in human hair, or that they are primary carriers of diseases, especially during the COVID-19 pandemic [10]. These misconceptions contribute to a general fear and misunderstanding of bats, overshadowing their ecological importance and unique biological features. Bats are ecologically vital, providing essential benefits such as pest control by consuming large quantities of insects, pollination of numerous plants, and seed dispersal that aids in forest regeneration. Their guano enriches soil fertility, contributing to nutrient cycling. These roles help maintain ecosystem balance and biodiversity [14].

In “What Is it Like to Be a Bat?”, Thomas Nagel [11] explores the unique, subjective experience of consciousness, using bats as a primary example due to their use of echolocation, a sensory perception vastly different from human experiences. He argues that understanding what it is like to be another creature is inherently limited by our inability to fully grasp their subjective experiences.

This concept directly inspires us to create EchoVision, a mixed reality interactive experience that challenges and transforms these misconceptions by providing participants with an immersive, empathetic experience of the world from a bat’s perspective. By simulating echolocation using sound visualization and mixed reality technology, EchoVision allows users to understand how bats navigate and perceive their environment with remarkable precision. By adjusting visual feedback based on the user’s voice, EchoVision bridges the gap between human and bat perception, offering a deeper, empathetic understanding of non-human sensory worlds and emphasizing the profound differences in how various species interact with their environments. This interactive experience highlights the sophistication and elegance of bat echolocation, fostering a greater appreciation for their role in ecosystems, such as pest control and pollination.

For maximum public exposure and impact in scientific education, we aim to go beyond traditional museum exhibits. We decided our premier installation would be showcased in a natural setting — the bat habitat under the Congress Avenue Bridge in Austin, Texas (“bat bridge” for short). This site attracts many tourists and locals who gather to watch millions of bats foraging after sunset. It is an ideal location to engage a large audience who may wish to learn more about bats. However, implementing mixed reality in this ad-hoc, in-the-wild setting presents challenges. Traditional mixed or virtual reality experiences often require expensive equipment and time-consuming setup [2, 13]. Usually, an assistant is needed to help audiences wear the devices. This inconvenience, combined with the time-consuming process, can lead to long waiting times and discourage participation.

To accommodate a large audience in-the-wild, we’ve created a custom-designed, handheld, bat-shaped mixed reality mask. This mask is based on the open-source HoloKit [8] project, an optical see-through mixed reality headset. The mask’s handheld handle

provides affordance for audiences to easily understand and engage with the experience, and then share it by passing the mask to the next user via the handle. The bat-shaped appearance piques the audience’s curiosity, encouraging them to try the experience. It also serves as an invitation to any uninformed passersby onsite.

Through EchoVision, participants can see bats not as menacing creatures but as fascinating and integral parts of the natural world. The installation encourages scientific learning and empathetic engagement, promoting an ecocentric design perspective that values the diverse ways non-human creatures interact with their surroundings. By offering a new perspective, EchoVision helps dispel myths, reduces unfounded fears, and inspires a deeper connection to and respect for bats and their vital ecological contributions.

2 RELATED WORK

2.1 Visualizing Echolocation

Bat echolocation is a sophisticated biological process that allows bats to navigate and hunt in the dark by emitting high-frequency sound waves and listening to the echoes that bounce back from objects in their environment. When a bat emits a sound, it travels through the air and reflects off surfaces, returning to the bat as an echo. By analyzing the time delay and changes in the frequency of the returning echoes, bats can determine the distance, size, shape, and texture of objects, as well as the speed and direction of moving prey. Teshima et al. [16] used precise scientific acoustic simulations to visualize the bat’s echolocation process. For educational purposes, research is also being conducted on immersive echolocation visualization in Virtual Reality [17].

2.2 MR for scientific education in exhibition

MR is increasingly being used in exhibitions and education [4]. It overlays digital content onto the user’s real-world view, providing an immersive interactive experience. This technology has revolutionized communication, breaking down physical barriers and simplifying learning [7]. In contexts such as field investigations, and outdoor education, it can facilitate innovative educational materials to simplify complex knowledge acquisition [1, 6]. Specifically, it can enhance interaction between participants and animals in their habitats without causing disturbance [2].

However, immersive MR is not very common in in-the-wild exhibitions and scientific education due to several obstacles. The cost of MR head-mounted displays can deter potential users, and challenges in user experience and ergonomics may discourage organizers from using MR [7, 15]. The performance of the device and the ergonomics of the MR headset greatly influence the user experience. Wearing an MR headset for a long time is uncomfortable, and the weight of the headset can cause discomfort during extended use [7]. Face contact pressure and muscle fatigue are also important evaluation criteria [5]. Increased weight can significantly reduce comfortable wear time [12] and cause pain and discomfort in the neck [3]. Furthermore, novice users’ unfamiliarity with new spatial interaction paradigms, like gesture input [9], can frustrate them during the experience. Therefore, despite its many benefits, the high barriers for beginners keep many users and organizers from adopting MR.

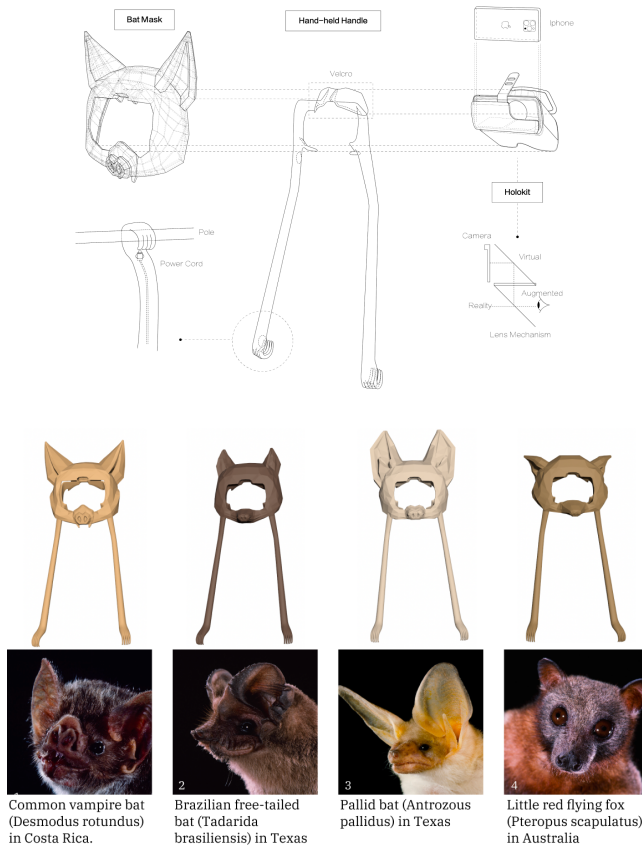


Figure 2: Bat-shaped Mask Design: (Top) Explosion drawings of the customized bat-shaped masks with the MR headset HoloKit X inside; (Middle) The design of the masks draws inspiration from four distinct species of bats, including two native to Austin; (Bottom) Photo credits for bat images: Merlin Tuttle’s Bat Conservation.

3 DESIGN

3.1 Designing the Bat-shaped Mask

The mask’s core is the HoloKit [8] headset, a smartphone-based mixed-reality optical see-through headset. The bat-shaped mask wraps around the HoloKit headset and attaches to its headpad with Velcro, replacing the original headband. An app running on the iPhone inside the HoloKit facilitates stereoscopic rendering. The design ensures that the camera and LiDAR sensor from the iPhone are not covered. Additionally, the optical see-through area of the headset is left open so the audience can see through it.

The handles, designed to resemble bat legs with claws, allow the mask to hang upside down from poles, ropes, or branches, mimicking the natural resting position of bats. The handles are detachable from the mask for easy storage and replacement. The design of the handles provides an affordance for the audience to easily understand how to manipulate the mask.

The appearance of masks draws inspiration from four distinct bat species, including two native to Austin: the Pallid bat and the

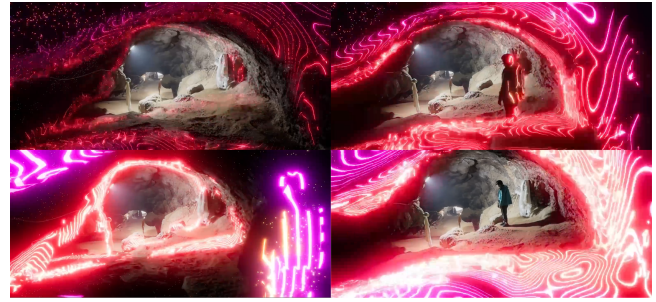


Figure 3: The visualized echolocation is simulated in the EchoVision MR experience using LiDAR and voice detection within the actual cave inhabited by bats.

Brazilian free-tailed bat. The masks’ color scheme mirrors the fur of these bats and uses a low-poly aesthetic to simplify their complex forms, highlighting the primary features of each species (Figure 2). The entire mask was modeled in Rhino, 3D printed in resin, and then meticulously painted to achieve the desired aesthetic.

3.2 Designing the Echolocation MR Experience

Leveraging the similarity between echolocation and LiDAR, EchoVision visualizes sound waves to mimic bat echolocation, as shown in Figure 3. It illustrates sound propagation in the environment reconstructed by the LiDAR sensor, artistically simulating the echolocation experience and drawing a parallel between human-made technology and natural biological processes. The visual effects of sound waves are primarily influenced by three key attributes: the volume of the user’s real-time voice input, the pitch of the voice, and the surrounding environment. The design details are as follows.

Sound Propagation and Brightness: The visualized sound waves propagate forward over time, and their behavior changes based on the volume of the voice. Louder voices result in a wider propagation angle and increased brightness of the wave lines. Users will be visually rewarded for trying different volume levels to see the various ways sound propagates.

Pitch and Wave Thickness: Higher-pitched voices travel farther and are represented by finer wave lines. This correlation helps demonstrate how pitch amplifies the reach of sound waves. The visual variations encourage users to try different pitch levels.

Particle Effects: Particle effects are designed to illustrate sound wave reflections. When sound waves hit an object’s surface, particles burst in the direction of the reflected wave. This feature provides an intuitive visual cue of the sound wave rebounding off the surface in the environment.

Human Body Recognition: We aimed to adhere to physical intuition while incorporating artistic enhancements for visual appeal. To encourage audience engagement, we designed unique golden color waves and an additional particle effect when sound waves propagate and collide with human bodies, distinguishing them from visual effects on other surfaces. This feature outlines the human form and encourages users to engage with their surroundings by shouting toward nearby individuals, enriching the interactive and entertaining experience.



Figure 4: The pop-up in-the-wild exhibition at the Congress Avenue Bridge in Austin, Texas; (a) (b) Bat-shaped masks were hung upside down on the sidewalk before the event; (c) More than 200 audience members and passersby gathered around the "bat bridge"; (d) The design of the mask handles allows easy passing to nearby people; (e) Multiple users can experience it simultaneously; (f) EchoVision exhibits are located under the "bat bridge," the habitat of bats.

4 THE POP-UP IN-THE-WILD EXHIBITION

We premiered EchoVision at The Congress Avenue Bridge in Austin, Texas, which famously hosts the largest urban bat colony in the world, with roughly 1.5 million Brazilian free-tailed bats residing beneath it every summer. Our performance started under the "bat bridge" with our collaborator, the bat conservationist Merlin Tuttle, who saved the millions of bats under the Austin bridges from extermination. During the event, more than 200 people listened to Merlin's captivating stories with bats, watched the huge number of bats flying out of the bat bridge after sunset, and experienced EchoVision to be like a bat and to have a rare glimpse into the bat's perceptual universe, as shown in Figure 4.

5 DISCUSSIONS

Rather than using mainstream solutions such as Quest or Vision Pro, we've customized the bat-shaped mask and transformed the HoloKit X into a handheld stereoscopic MR headset. Based on our experience with the pop-up exhibition, we've identified the following benefits of our handheld design for a large audience in ad-hoc, in-the-wild settings.

An Invitation to Spark Curiosity: The mask is designed to hang upside down like a real mask, catching the attention of passersby. The unique bat-shaped mask serves as an invitation to passersby,

sparking their curiosity to imagine the yet-unseen MR experience and encouraging interaction with the experience.

Affordance: Unlike traditional, bulky MR headsets that can be cumbersome and difficult to put on, the handheld device with handles is designed for quick and easy use. It inherently offers greater ease of use, allowing users to understand, grasp, and begin using the device with minimal instruction. This invites participation and helps bridge the gap often created by unfamiliar technology, reducing the sense of alienation that usually comes with high-tech interfaces.

Ease of Sharing and Self-Propagation: The handheld device can be easily passed from one user to another with minimal guidance. This efficiency has been proven in high-volume pop-up exhibitions, allowing us to engage over 200 participants quickly. The design also benefits from self-promotion, as passersby are often inspired to try it themselves after watching others. The handles on the mask simplify its use and transfer, fostering the natural spread of the experience, as shown in Figure 4 (d). This process conserves the organizers' manpower and enhances the social interaction throughout the exhibition.

Weight: Unlike traditional MR headsets, our handheld design avoids the discomfort of wearing a heavy device by offloading the weight to a comfortable handheld position, ensuring a more user-friendly experience.

No Isolation: The semi-open design of the handheld device allows users to easily disengage and re-engage with the MR experience. Reality is always in their field of vision through the optical see-through design, eliminating the sense of isolation often associated with VR. This feature also reduces dizziness and isolation, providing a safer, more comfortable experience.

Hygiene: The handheld design of the device addresses hygiene concerns to some extent. Users can maintain a gentle contact distance between the mask and their face by holding the handles. This is especially important in public pop-up settings, as it avoids disrupting makeup and hairstyles of the user's personal image like traditional wearable MR did, and lowers the barrier to entry.

Inclusive Engagement: Unlike traditional MR experiences that may bore non-participants, EchoVision keeps everyone engaged, involving both users and passersby. Passersby can photograph users, while users can try echolocating the human body by making sounds towards passersby. This creates a dynamic, inclusive, and entertaining atmosphere.

The handheld, bat-shaped MR mask design significantly improves usability and inclusivity, particularly in ad-hoc and public settings. This design is especially advantageous in crowded environments, where quick transitions between participants are essential to maintain engagement and flow. This MR project has addressed many of the drawbacks of existing MR headsets, such as headset weight, wearing inconvenience, and the feeling of detachment.

6 CONCLUSION

EchoVision allows participants to experience the *umwelt* of bats, promoting scientific learning, empathy, and an ecocentric design perspective that values non-human interactions. EchoVision dispels myths, reduces fears, and inspires respect for bats and their ecological contributions.

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